```
Nigeria, step 2 (2026-11-02)

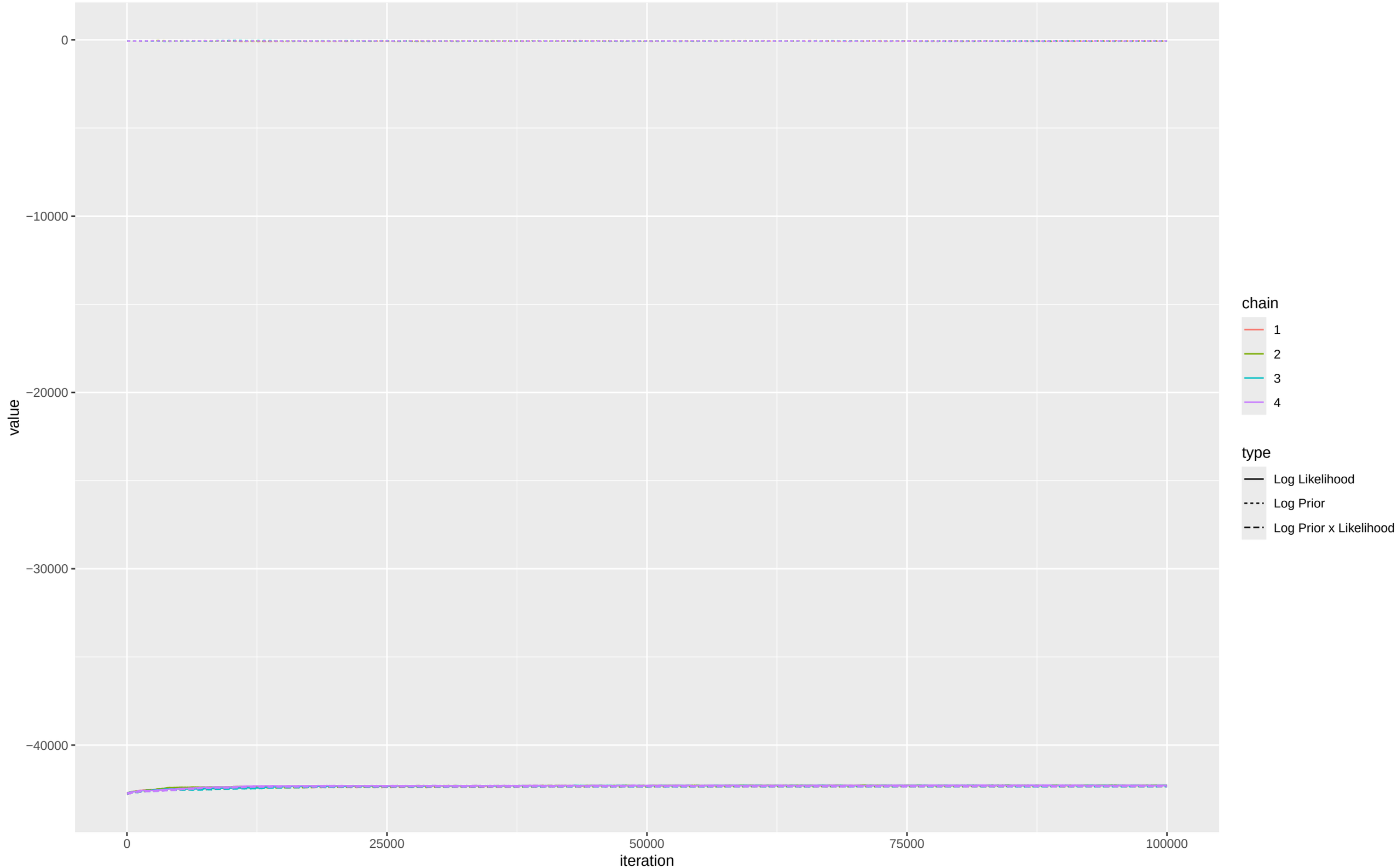
 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.279             -42289.1            -42276.3                 0
     2 20263017             5000           0.277             -42276.2            -42265.7                 0
     3 20263018             5000           0.276             -42285.7            -42269.8                 0
     4 20263019             5000           0.277             -42288.4            -42265.2                 0
                                    file
 mcmc_nigeria_step2_chain1_2026-10-31.Rdata
 mcmc_nigeria_step2_chain2_2026-10-31.Rdata
 mcmc_nigeria_step2_chain3_2026-11-01.Rdata
 mcmc_nigeria_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                                trate.0   age.25.to.49.hiv.mortality.multiplier.0    age.30.to.39.transmission.multiplier.1
                                  2.073                                     2.023                                     1.976
   age.15.to.19.transmission.multiplier.3    age.15.to.19.transmission.multiplier.0          male.hiv.mortality.multiplier.0
                                  1.837                                     1.686                                     1.633
                                trate.4                                   trate.3    age.0.to.14.hiv.mortality.multiplier.1
                                  1.594                                     1.563                                     1.531
                                trate.1               female.to.male.multiplier    age.20.to.29.transmission.multiplier.0
                                  1.431                                     1.419                                     1.399
   age.20.to.29.transmission.multiplier.1   age.15.to.24.hiv.mortality.multiplier.0    age.30.to.39.transmission.multiplier.0
                                  1.397                                     1.394                                     1.371
   age.40.to.49.transmission.multiplier.0    age.40.to.49.transmission.multiplier.2 age.50.and.over.transmission.multiplier.3
                                  1.364                                     1.362                                     1.327
          hiv.specific.mortality.rates.2    age.30.to.39.transmission.multiplier.3    age.0.to.14.hiv.mortality.multiplier.2
                                  1.320                                     1.287                                     1.286
   age.20.to.29.transmission.multiplier.3       over.50.hiv.mortality.multiplier.1       over.50.hiv.mortality.multiplier.2
                                  1.273                                     1.261                                     1.256
   age.40.to.49.transmission.multiplier.3
                                  1.245

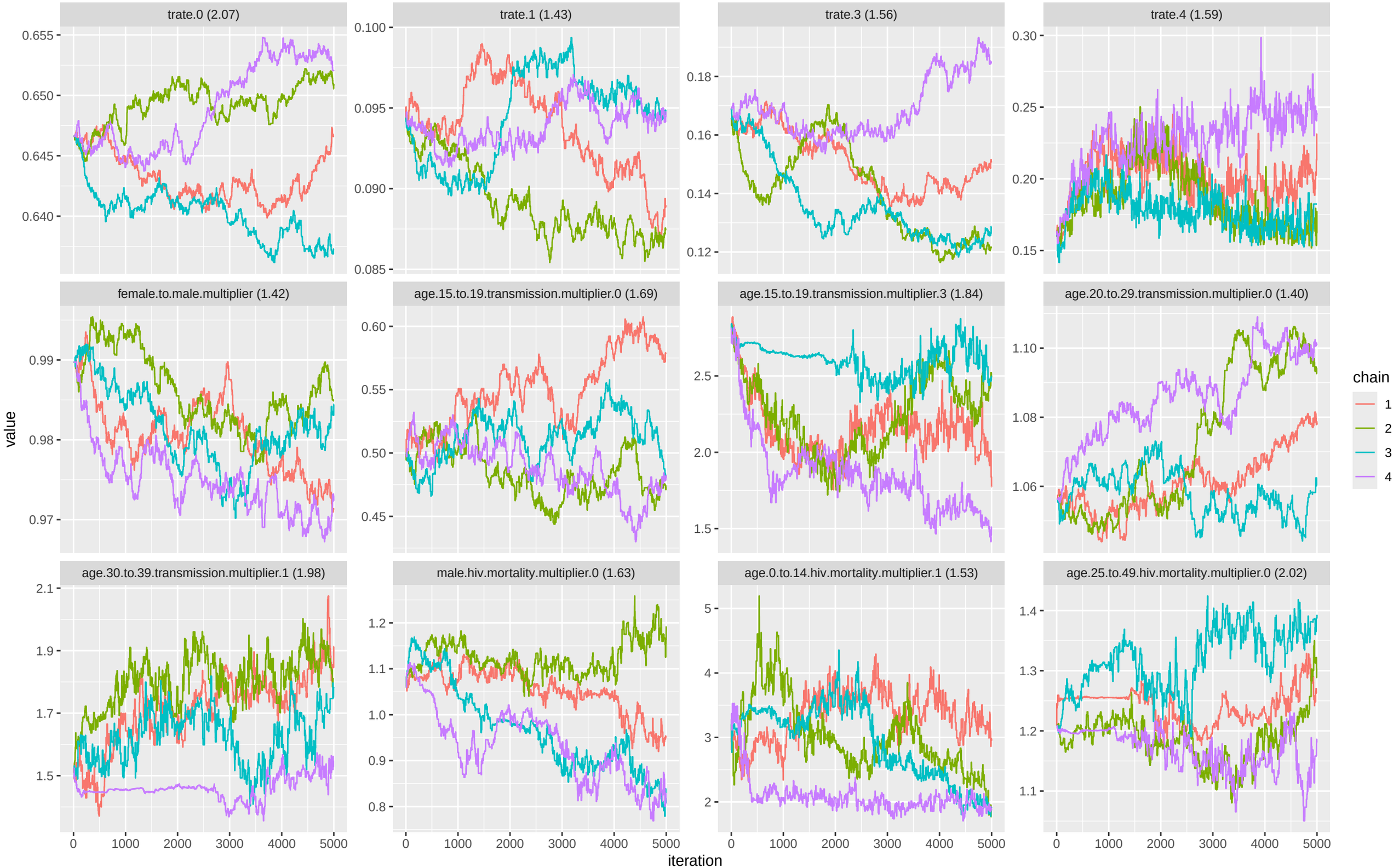
Merged simset:
 location simulations total.weight                                        file
  Nigeria        1000         1000 simset_nigeria_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 59 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       15-24 female              26                  23              x3.05      2025            x3.39
 hiv.mortality       10-19 female              26                   0              x2.76      2025            x3.51
 hiv.mortality       25-49   male              26                   0              x0.36      2025            x0.17
 hiv.mortality 50 and over female              26                   0              x2.34      2025            x1.37
 hiv.mortality       15-24   both              26                  12              x2.38      2025            x2.37
 hiv.mortality       25-49 female              26                   0              x0.44      2025            x0.33
 hiv.mortality       25-49   both              26                   0              x0.45      2025            x0.27
 hiv.mortality       10-19   both              26                   0              x2.15      2025            x2.30
 hiv.mortality 50 and over   both              26                   0              x2.06      2025            x1.56
     awareness        0-14   both              11                   0          -29.3 pts      2025        -29.3 pts
 hiv.mortality 50 and over   male              26                   8              x1.85      2025            x1.72
     incidence       10-19   male              26                   8              x0.70      2025            x0.67
    prevalence 50 and over   male              26                  19              x0.71      2025            x0.98
 hiv.mortality       15-24   male              26                  42              x1.45      2025            x0.81
     incidence       15-24   male              26                   8              x0.75      2025            x0.78
   suppression        0-14   both               7                   0          +11.5 pts      2025        +11.6 pts
 hiv.mortality   All ages   male              26                   8              x0.80      2025            x0.78
```

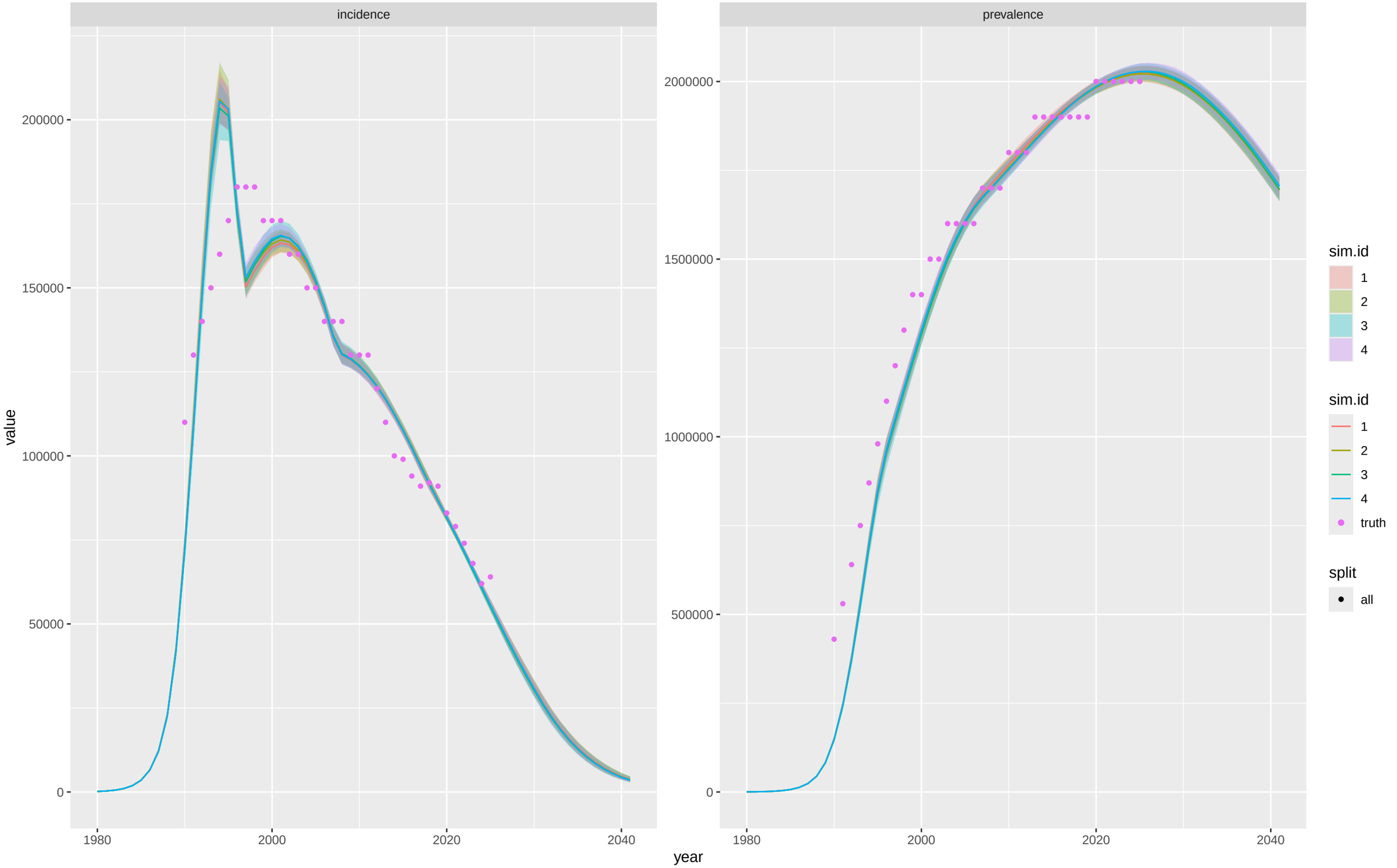
Nigeria step 2: likelihood



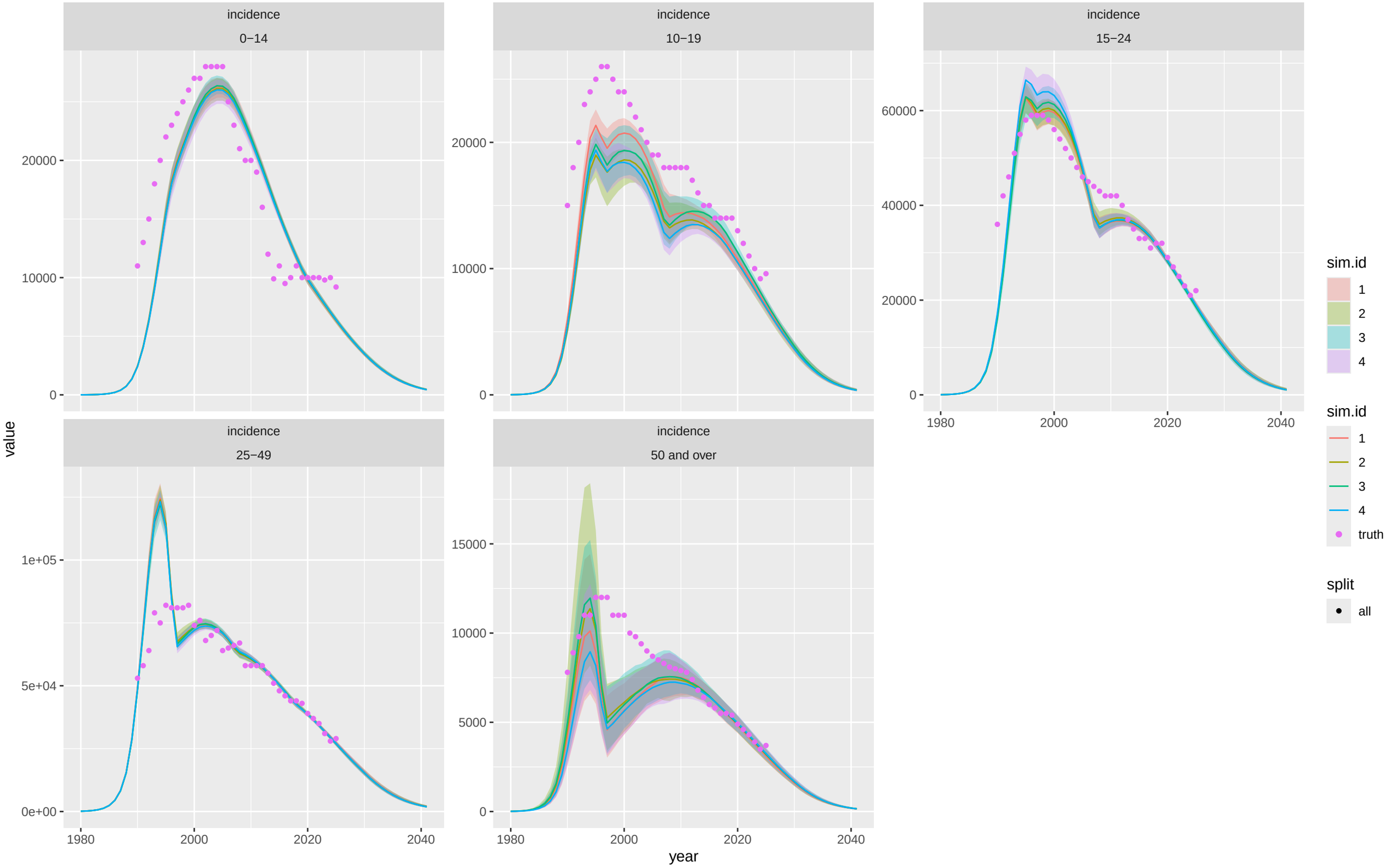
Nigeria step 2: trace



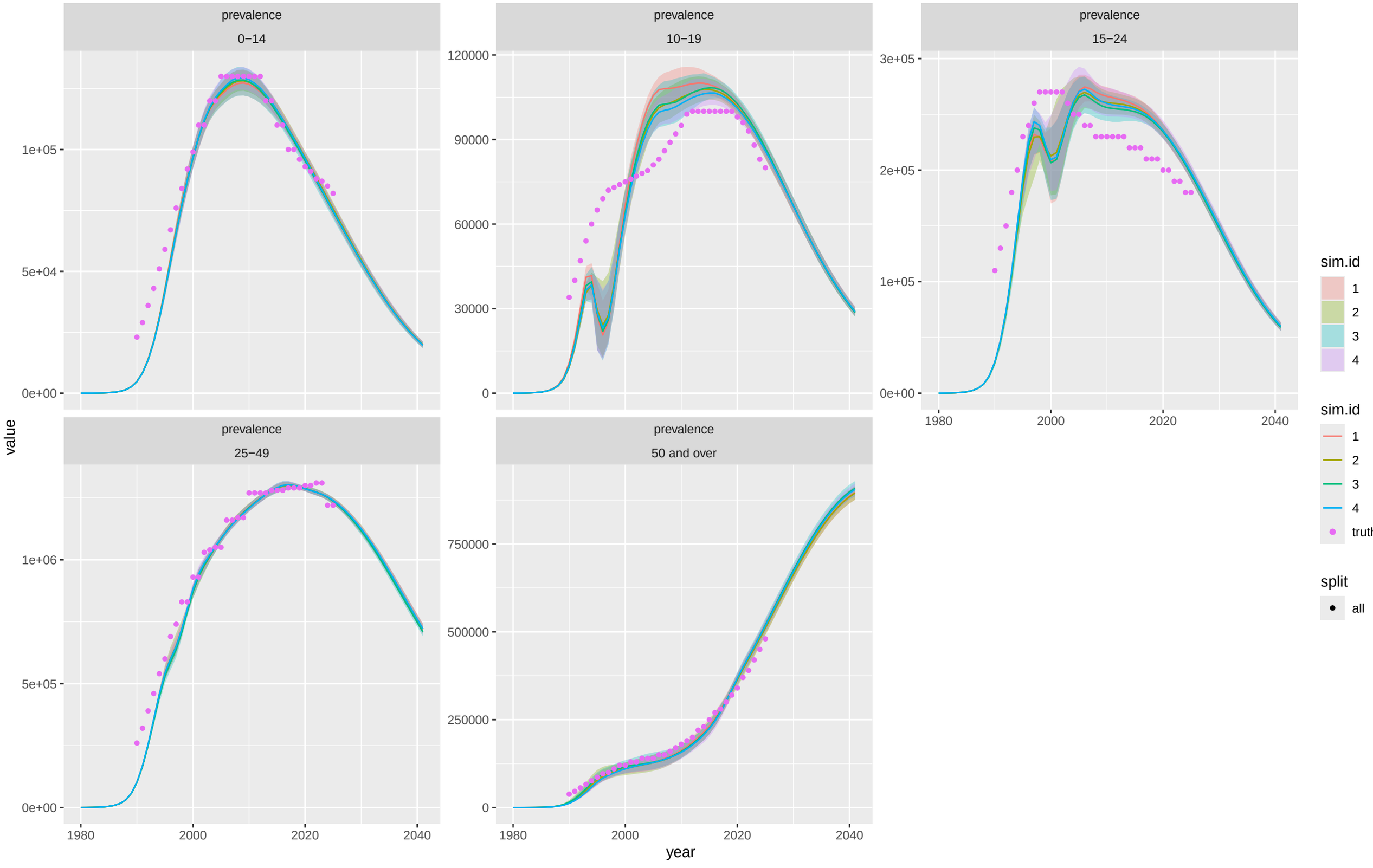
Nigeria step 2: fit (one color per chain)



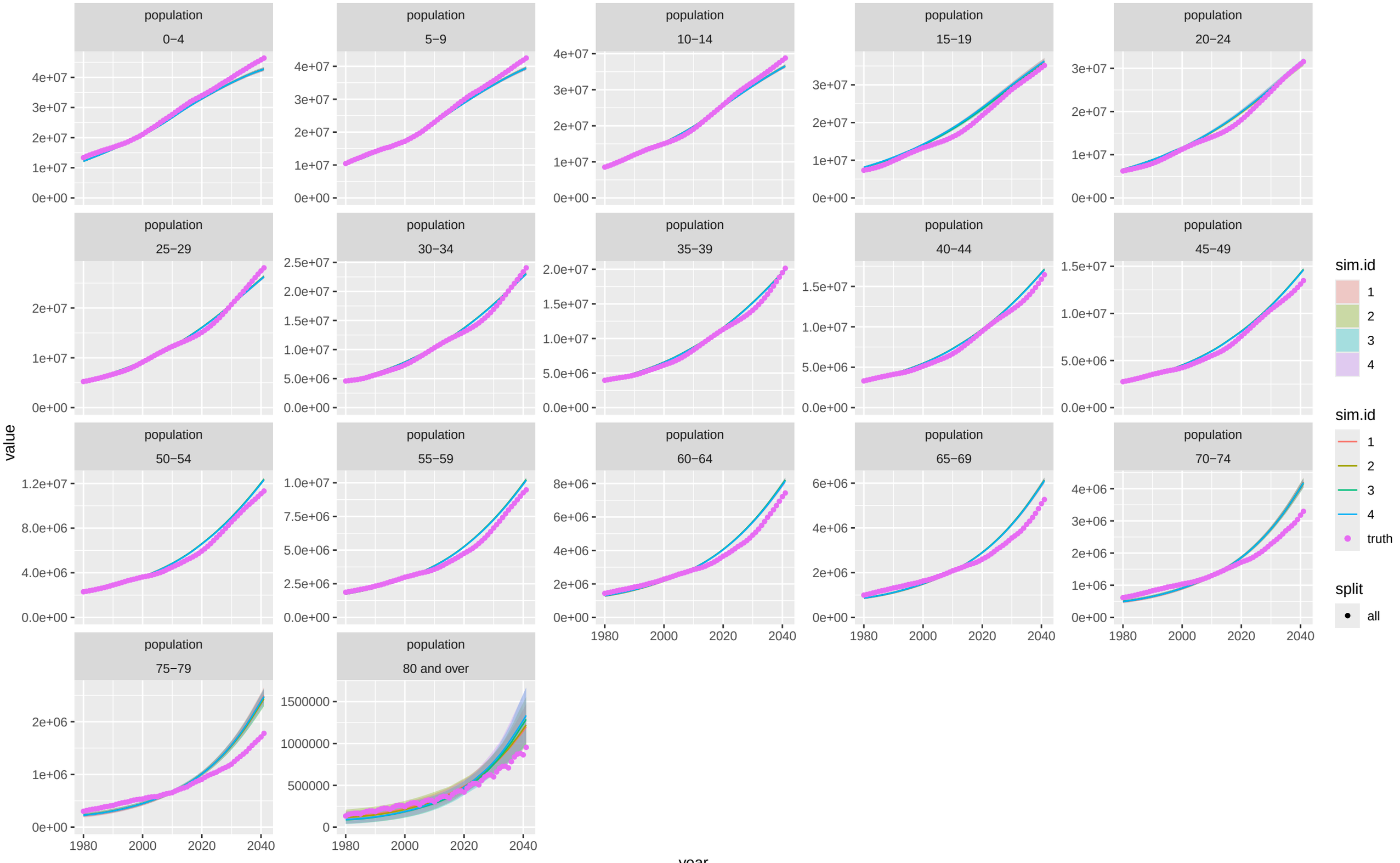
Nigeria step 2: incidence.by.age (one color per chain)



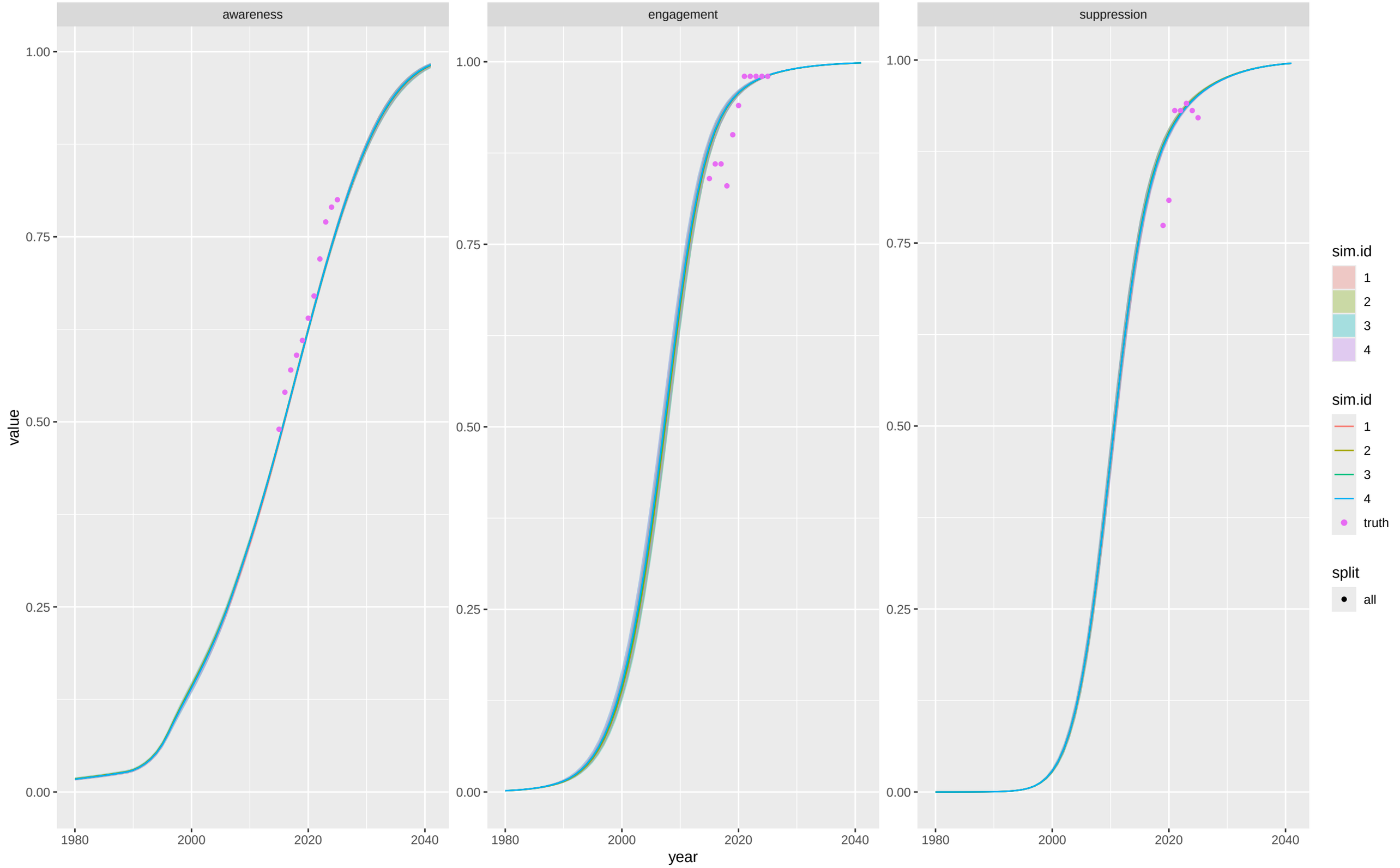
Nigeria step 2: prevalence.by.age (one color per chain)



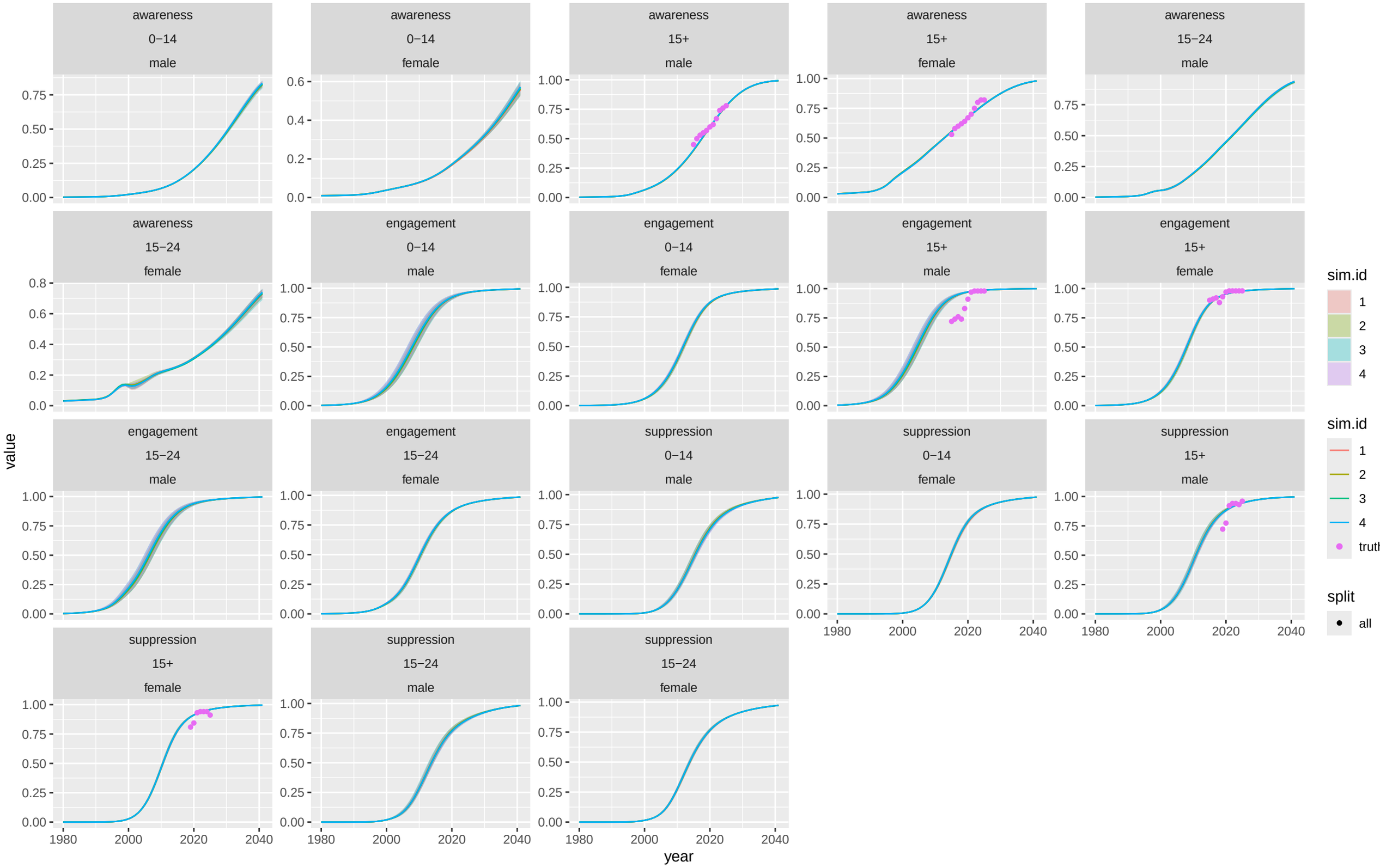
Nigeria step 2: population.by.age (one color per chain)



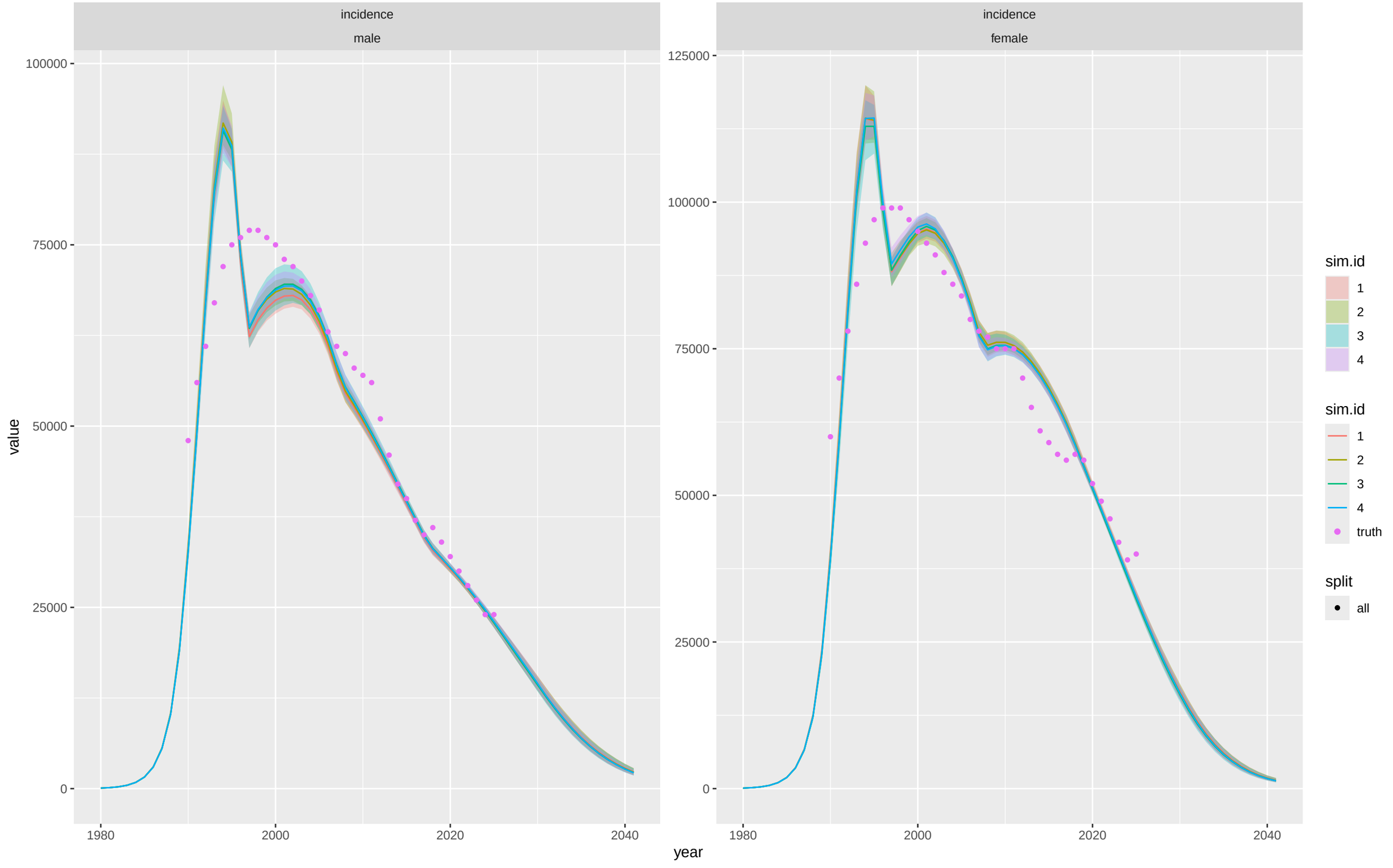
Nigeria step 2: cascade (one color per chain)



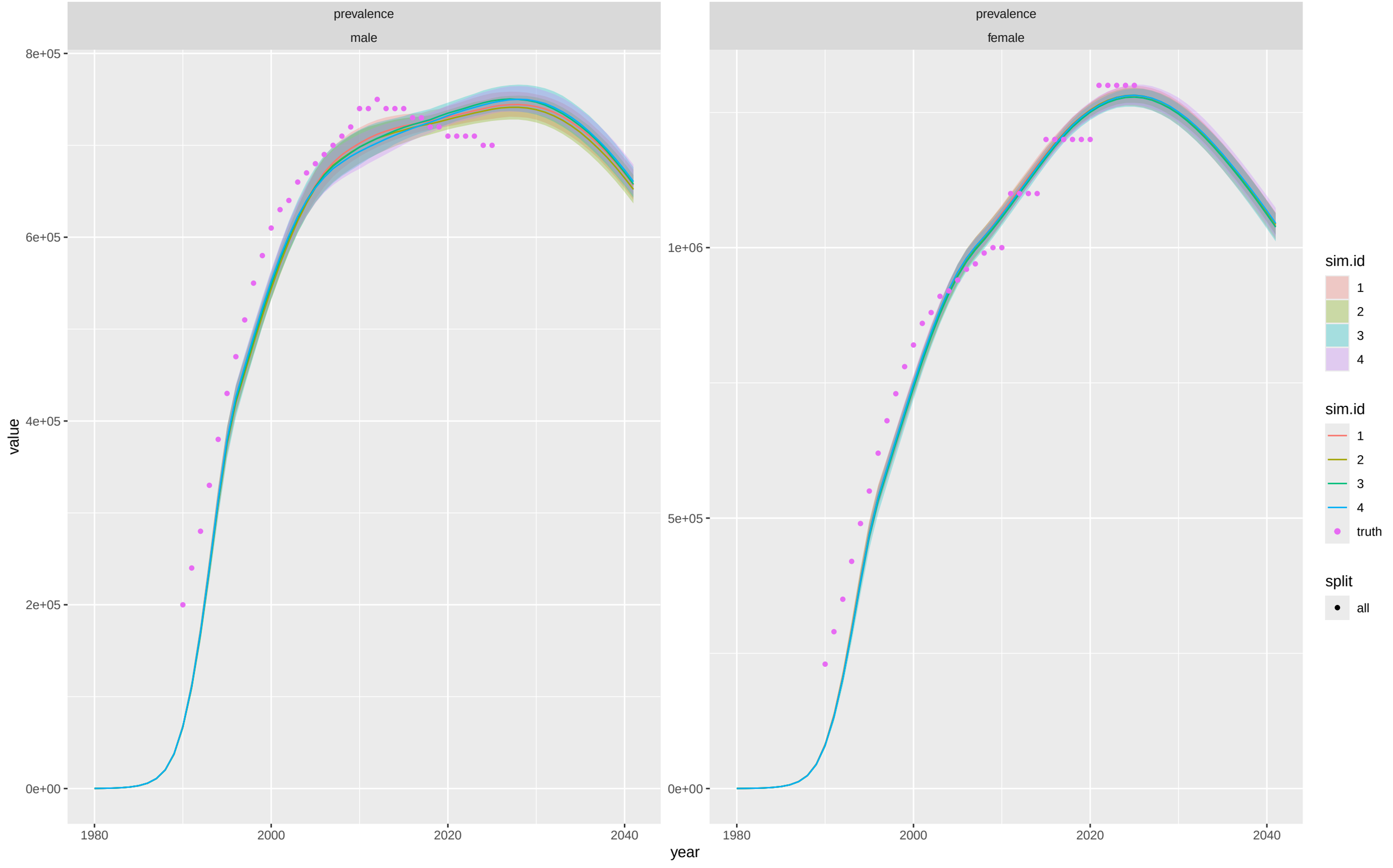
Nigeria step 2: cascade.by.age.sex (one color per chain)



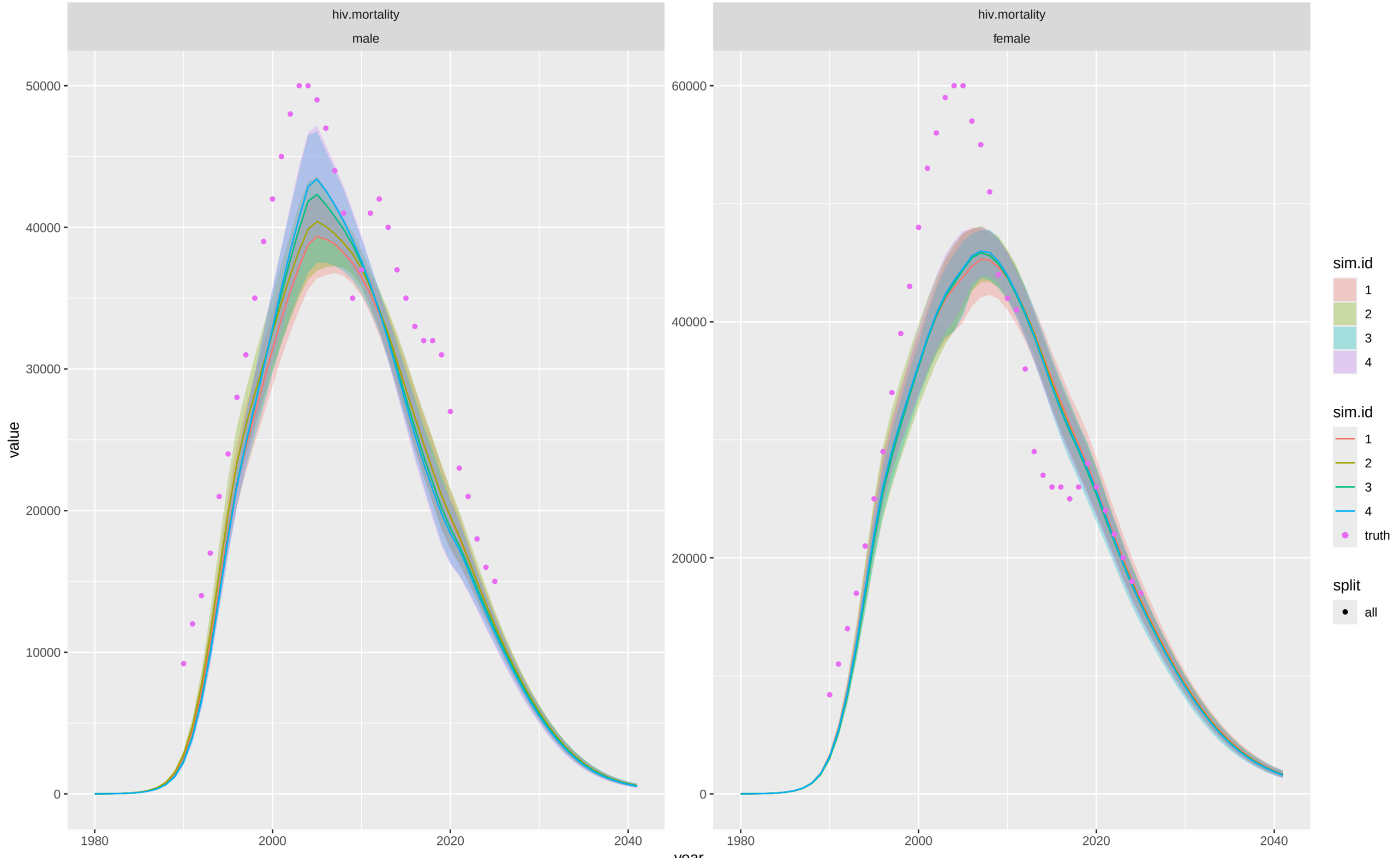
Nigeria step 2: incidence.by.sex (one color per chain)



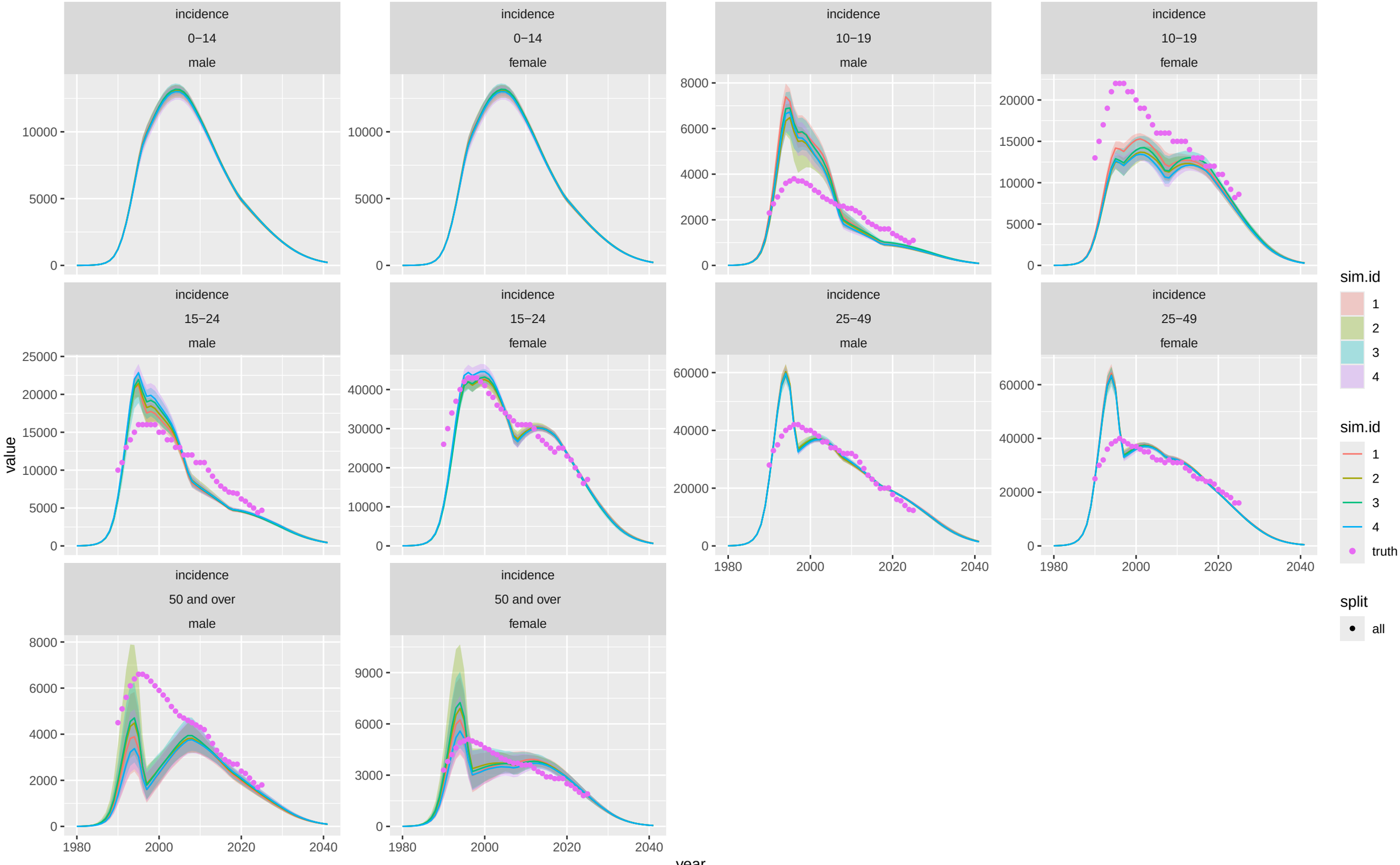
Nigeria step 2: prevalence.by.sex (one color per chain)



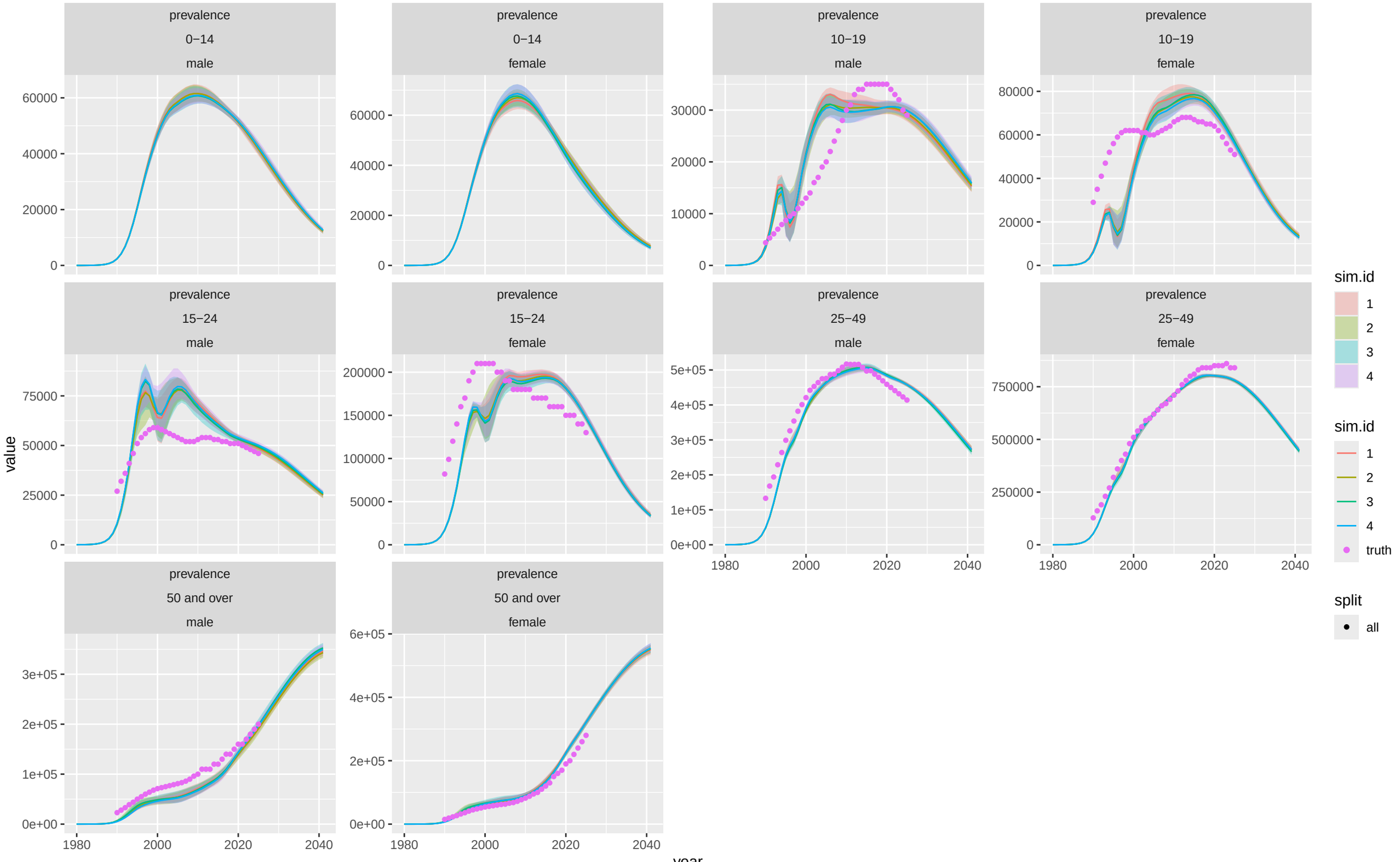
Nigeria step 2: hiv.mortality.by.sex (one color per chain)



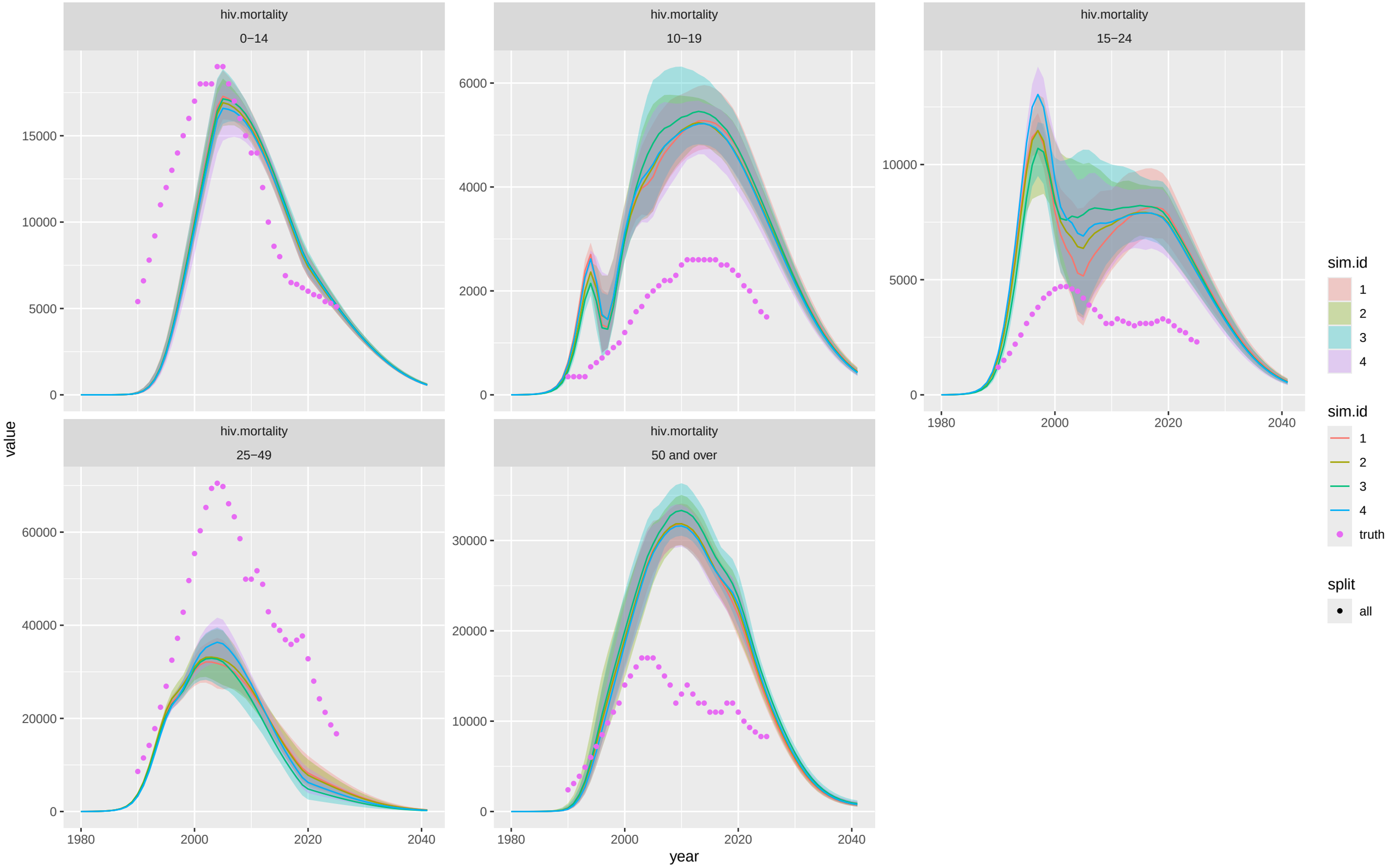
Nigeria step 2: incidence.by.age.sex (one color per chain)



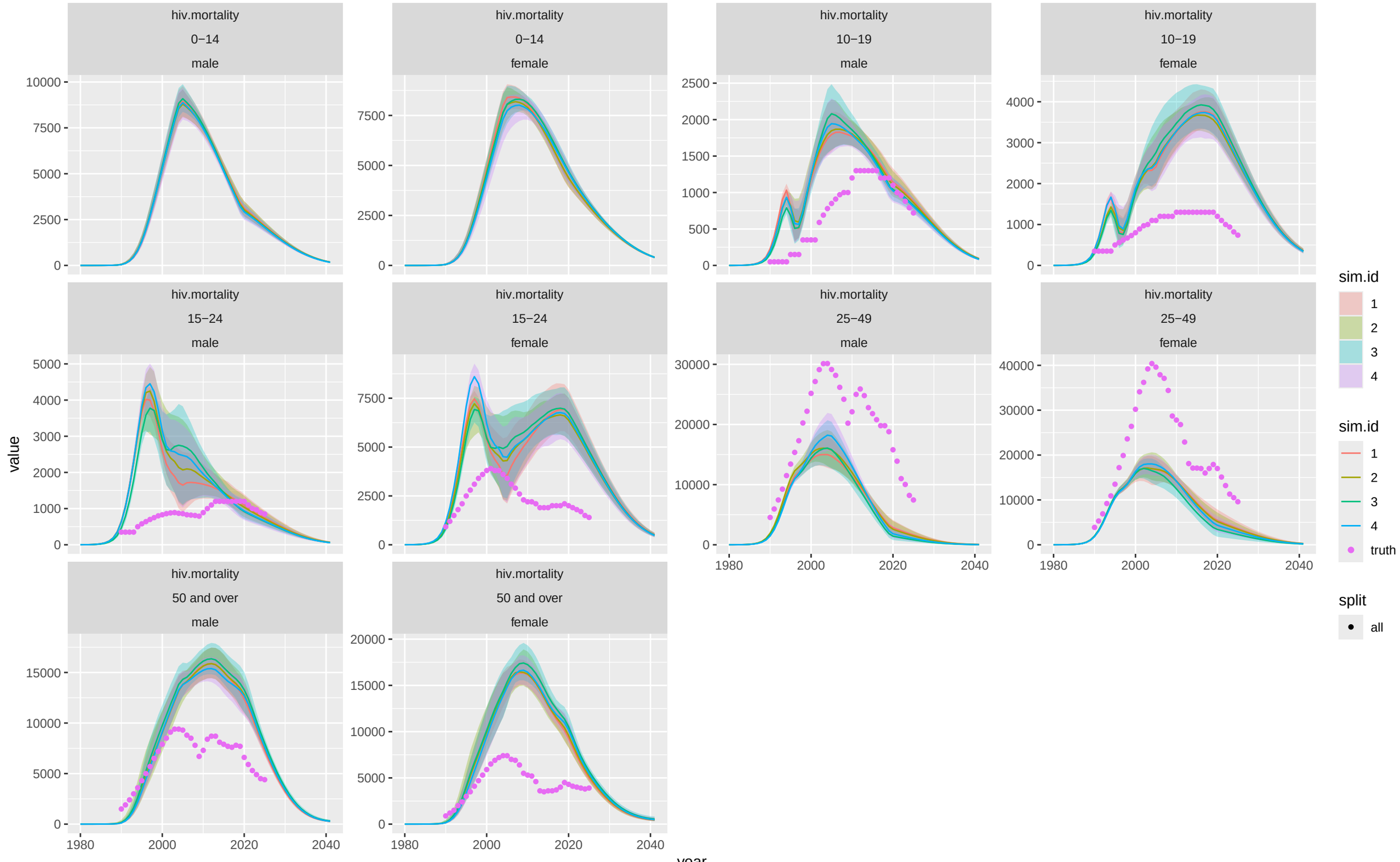
Nigeria step 2: prevalence.by.age.sex (one color per chain)



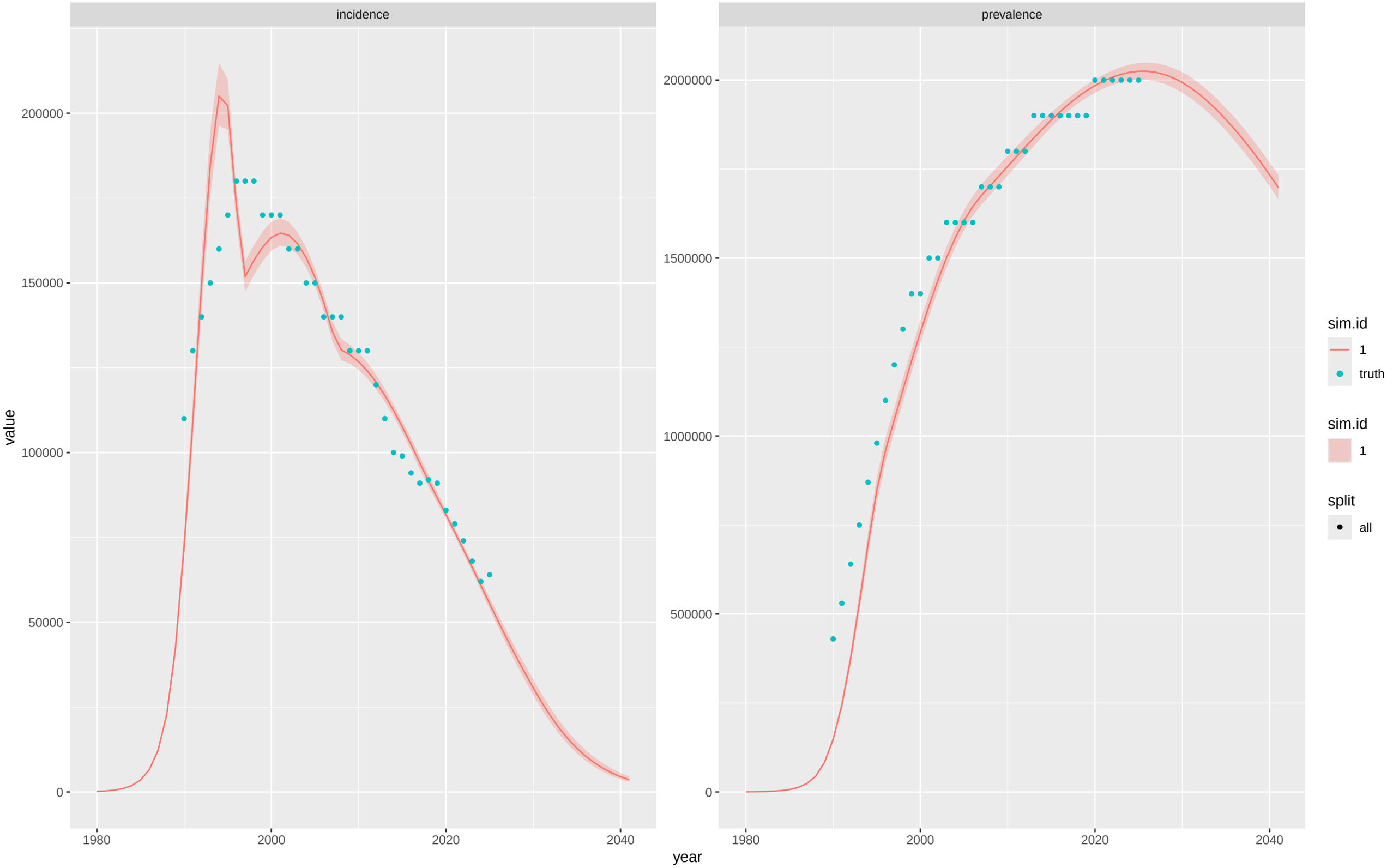
Nigeria step 2: hiv.mortality.by.age (one color per chain)



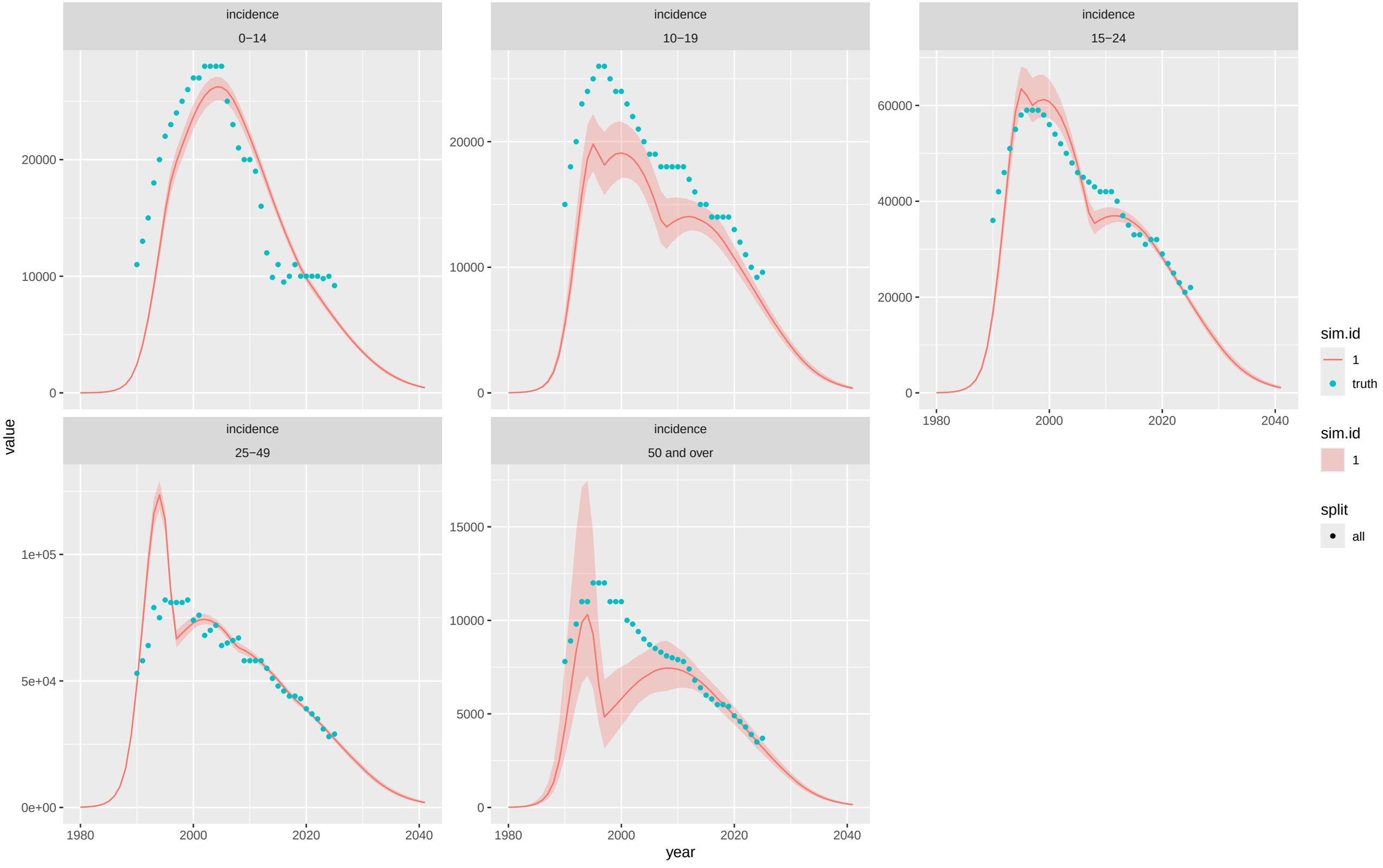
Nigeria step 2: hiv.mortality.by.age.sex (one color per chain)



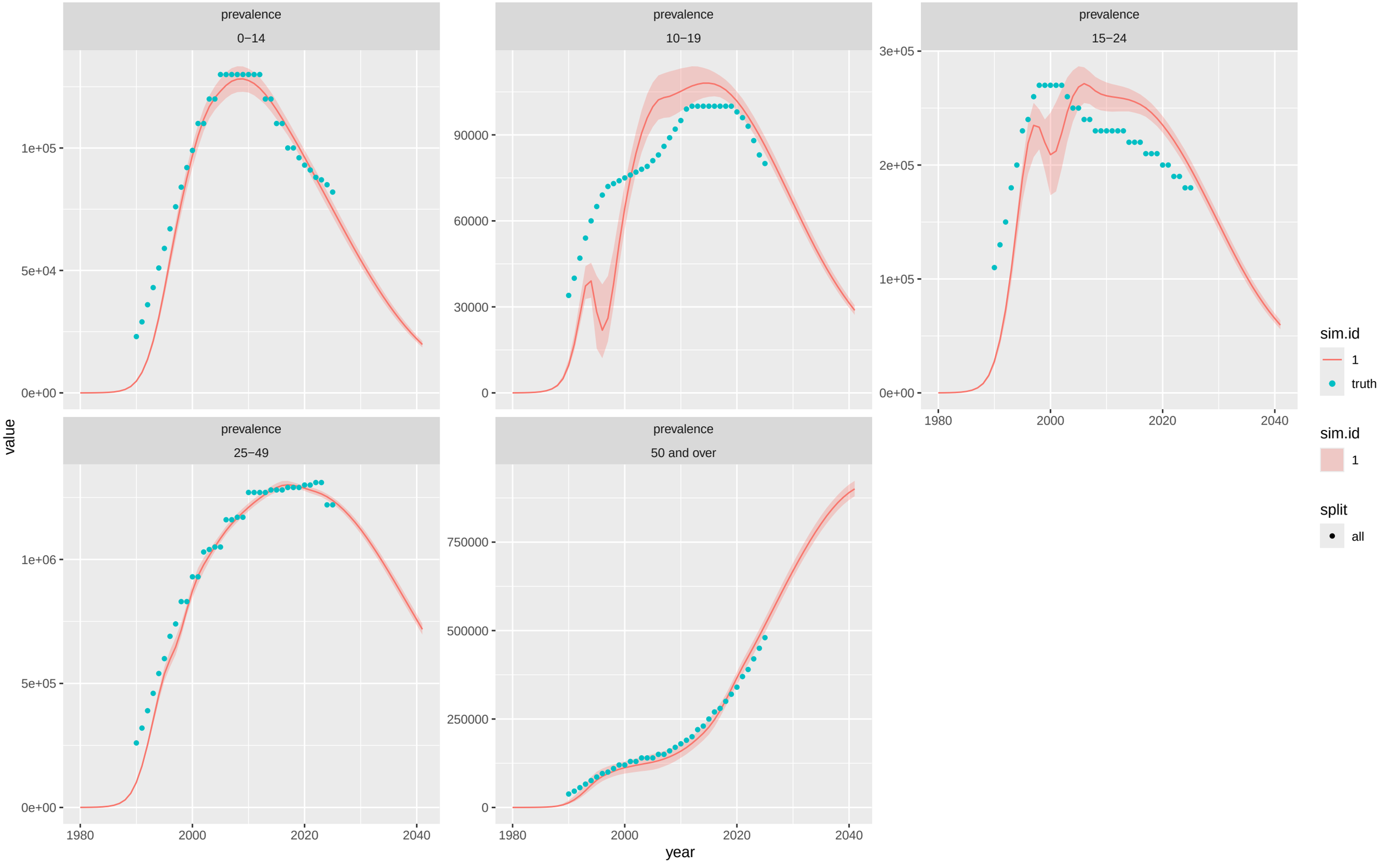
Nigeria: fit (merged Step 2 simset)



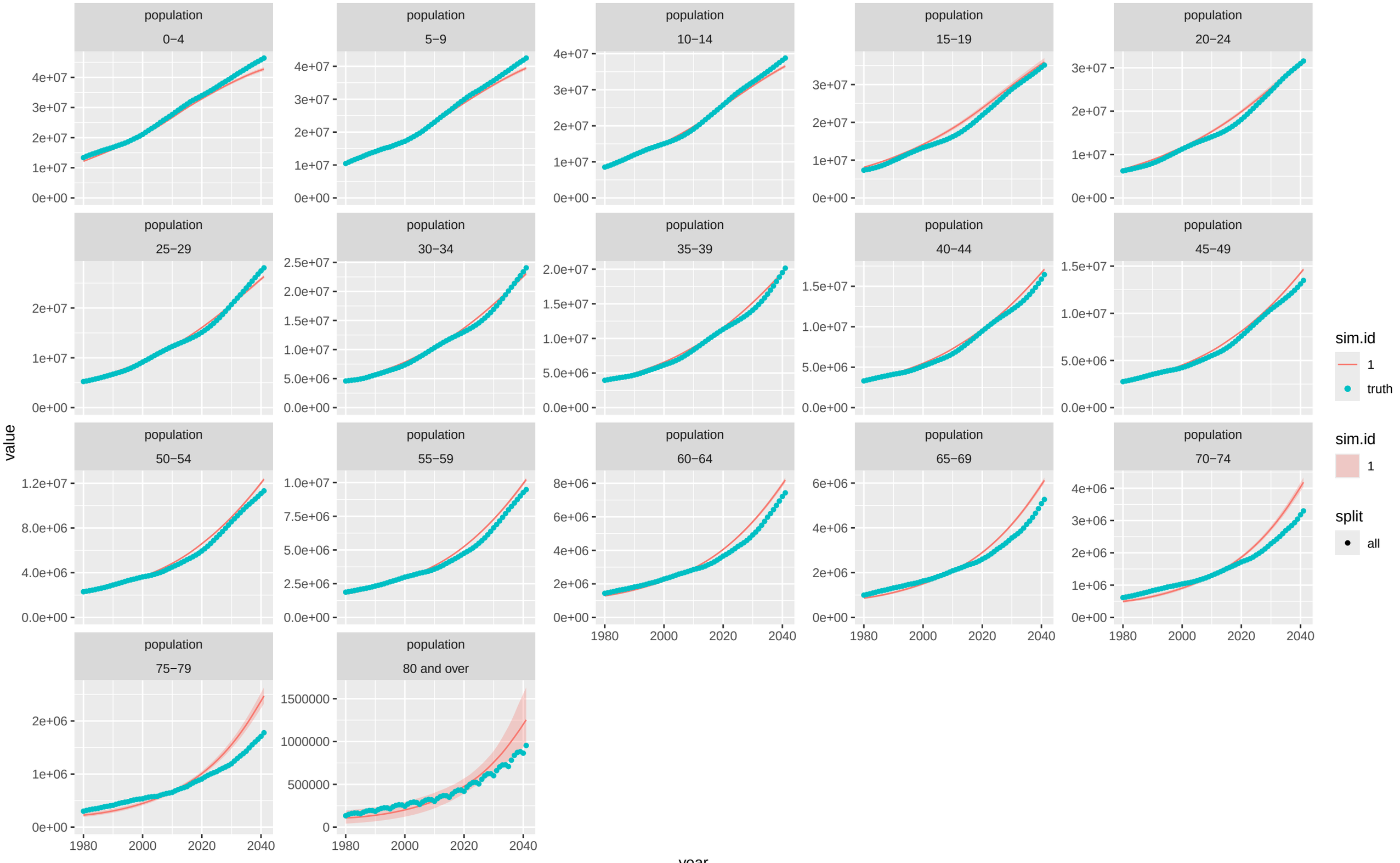
Nigeria: incidence.by.age (merged Step 2 simset)



Nigeria: prevalence.by.age (merged Step 2 simset)



Nigeria: population.by.age (merged Step 2 simset)



Nigeria: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

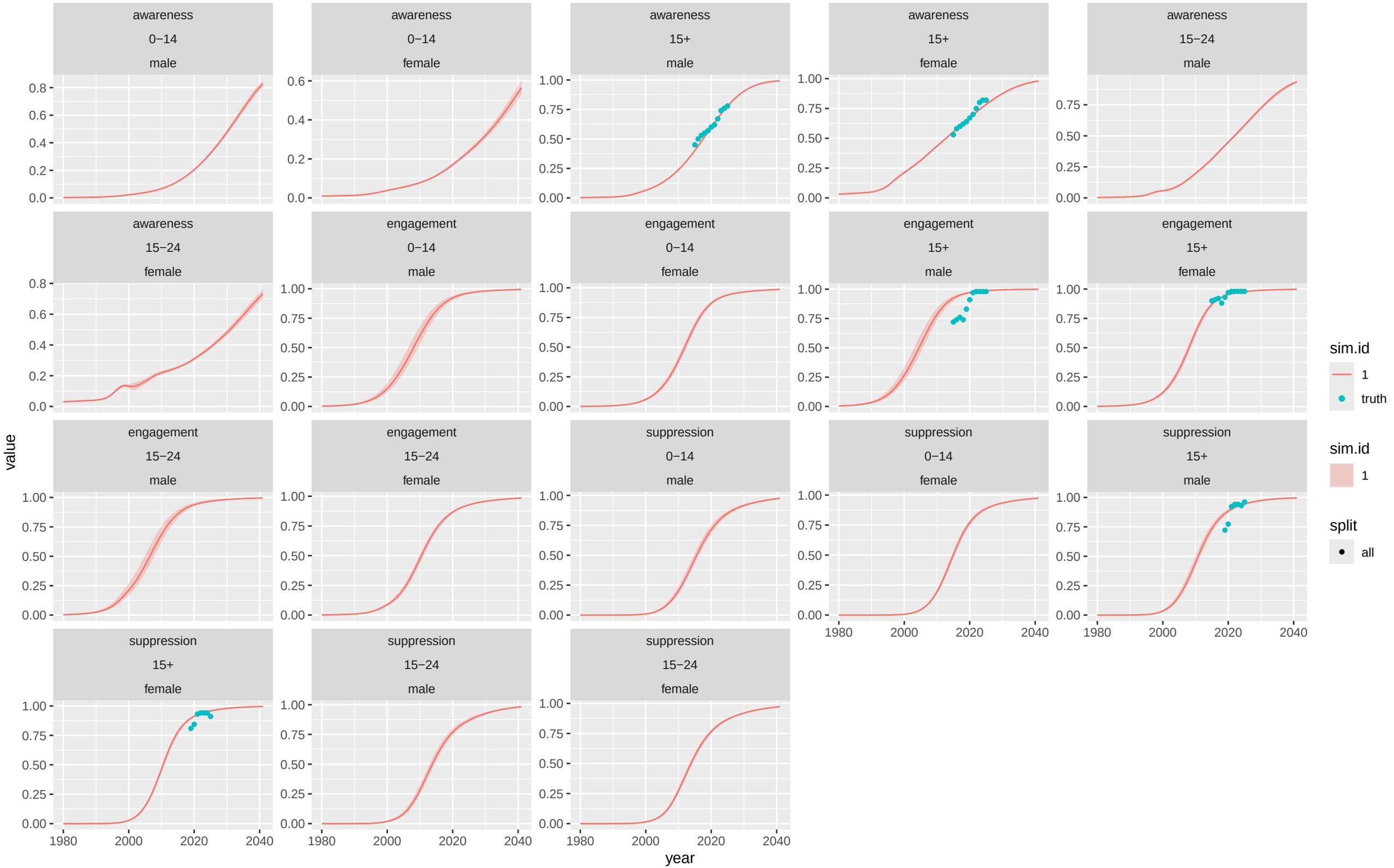
sim.id

1

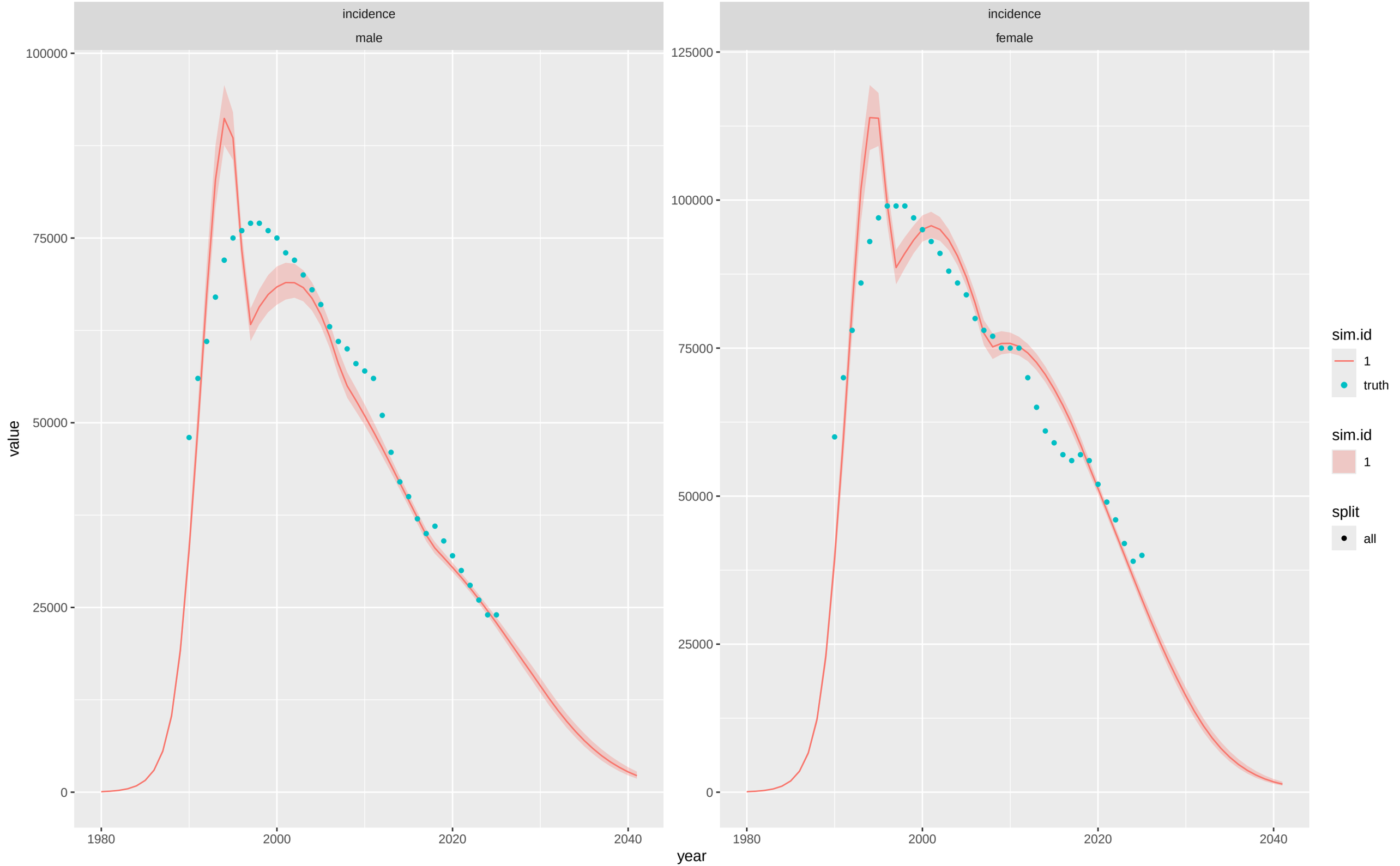
split

all

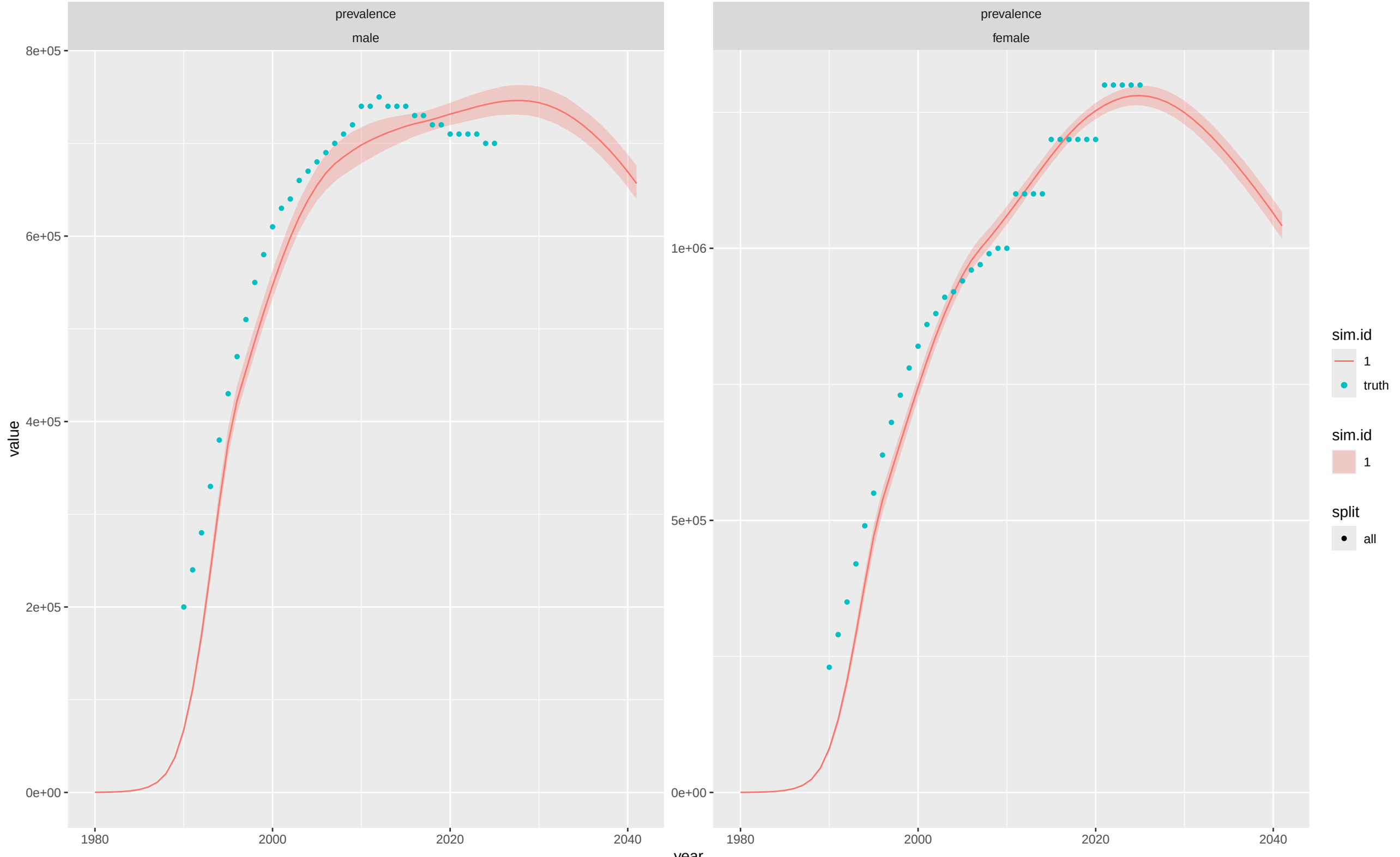
Nigeria: cascade.by.age.sex (merged Step 2 simset)



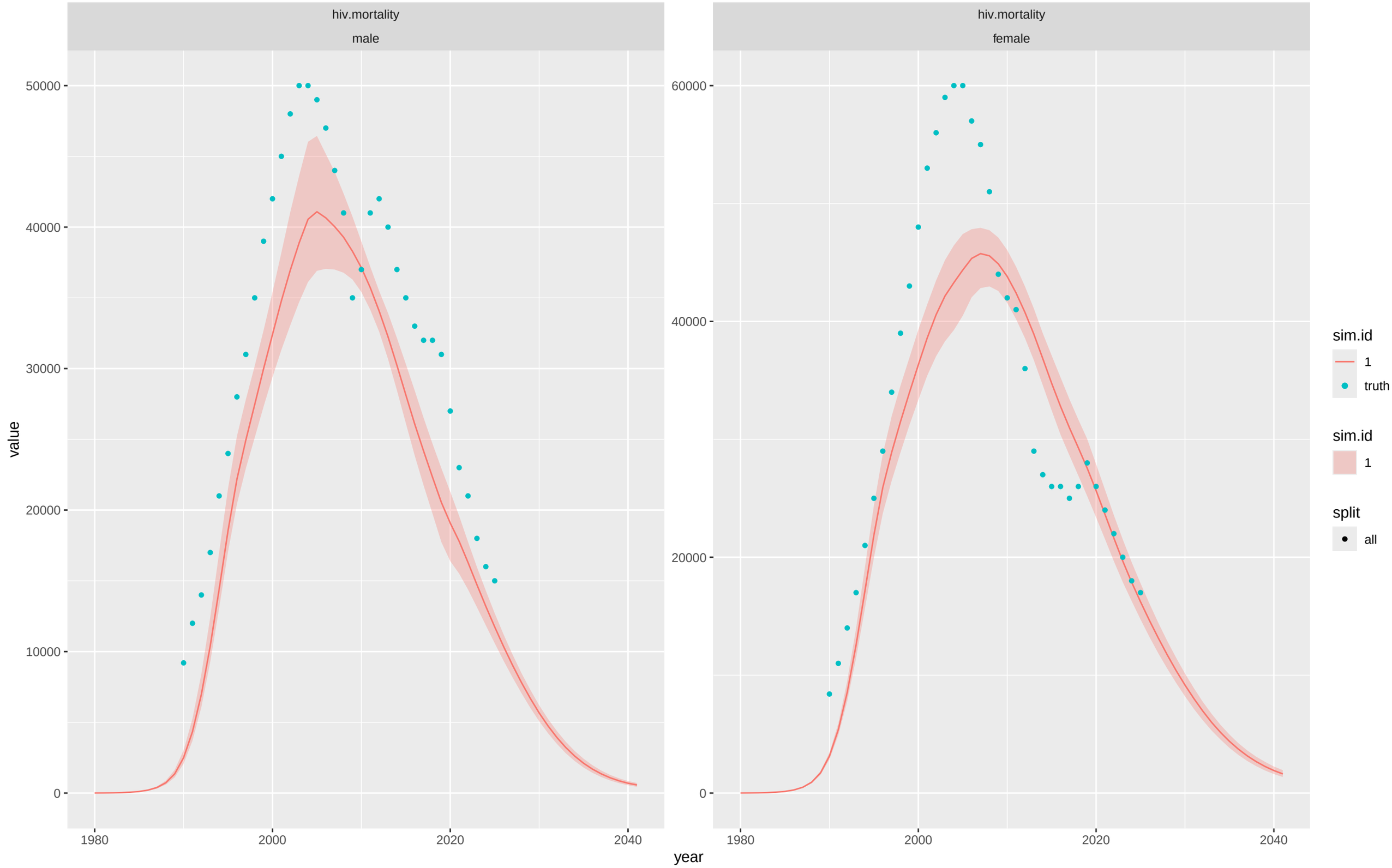
Nigeria: incidence.by.sex (merged Step 2 simset)



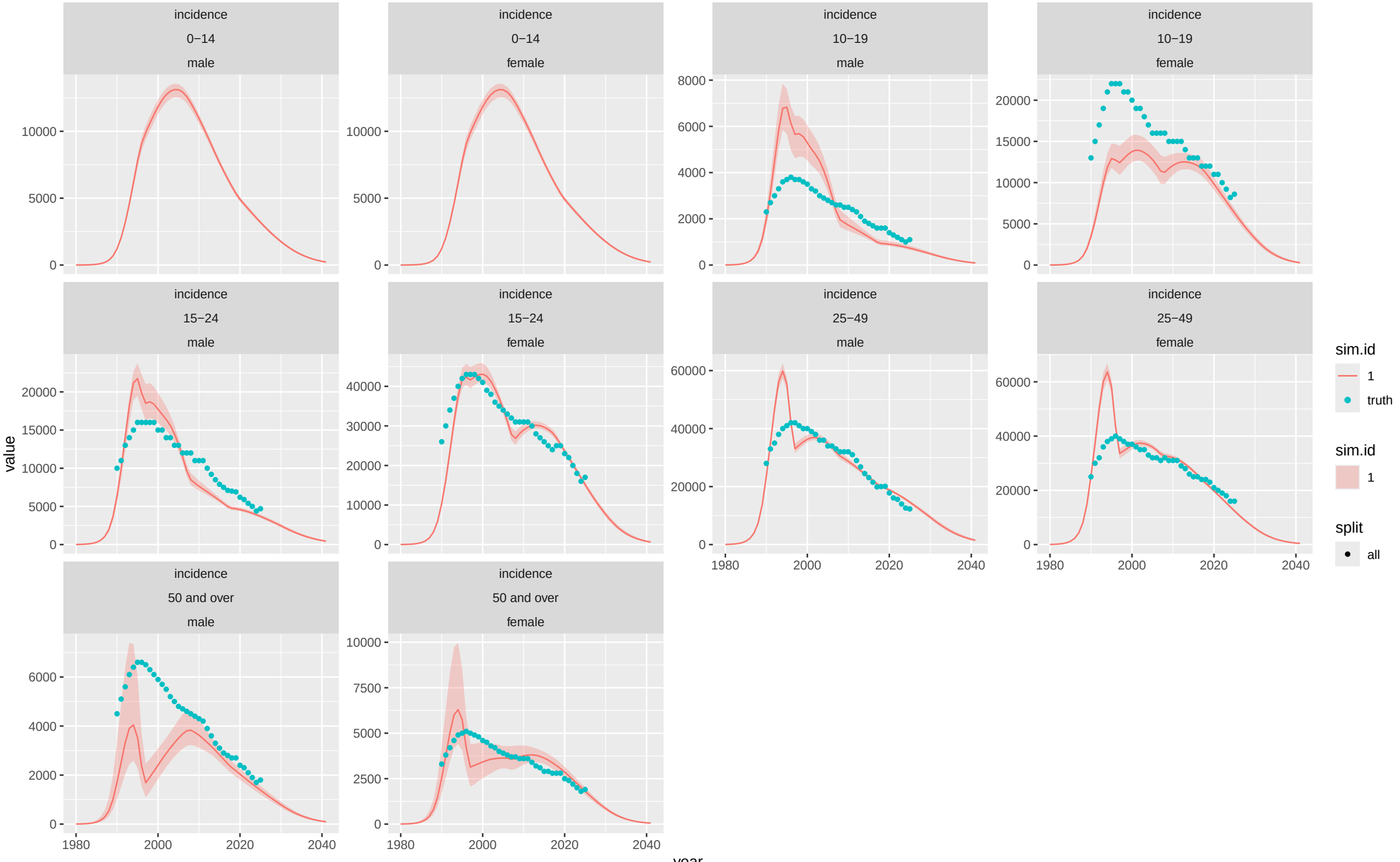
Nigeria: prevalence.by.sex (merged Step 2 simset)



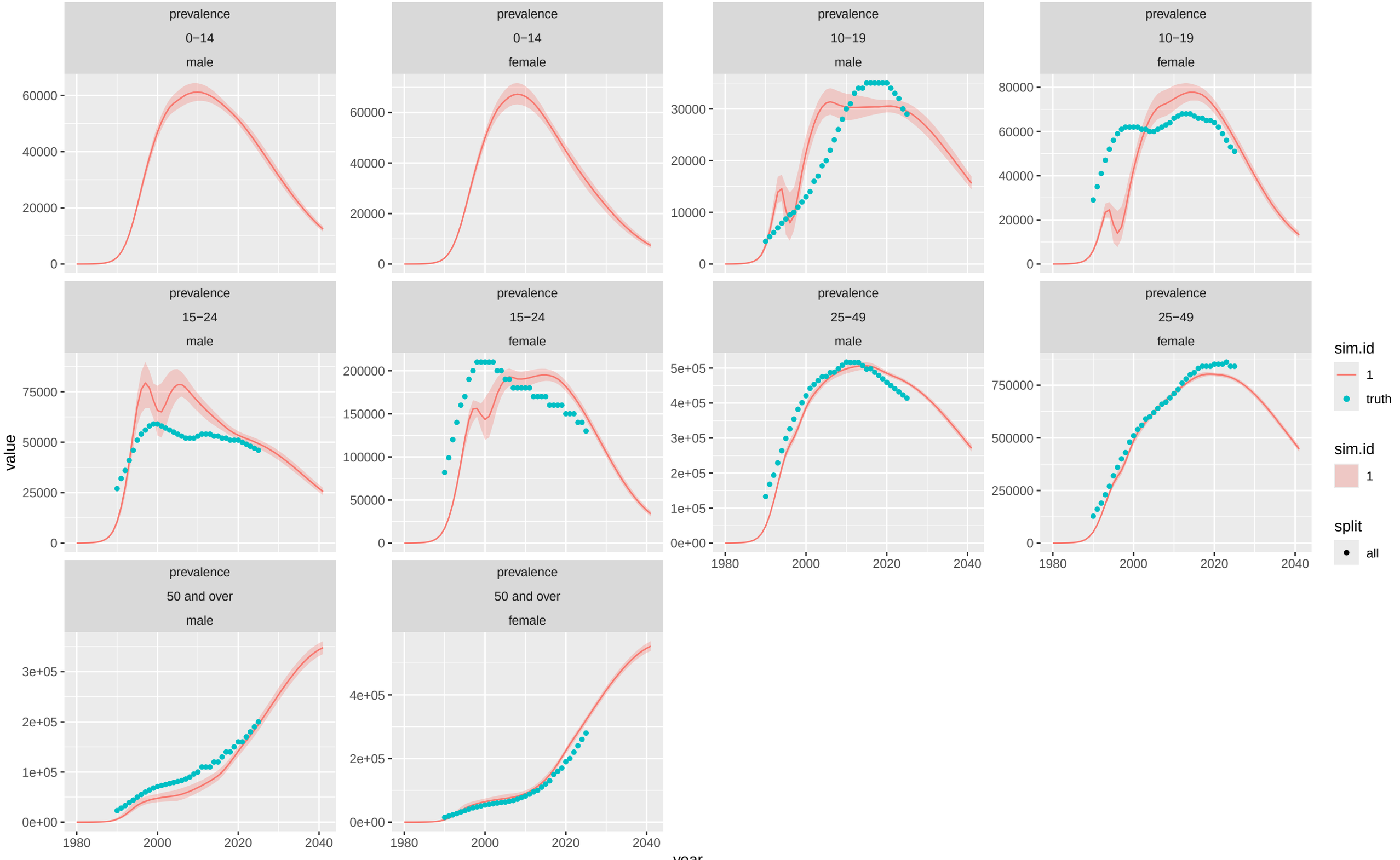
Nigeria: hiv.mortality.by.sex (merged Step 2 simset)



Nigeria: incidence.by.age.sex (merged Step 2 simset)



Nigeria: prevalence.by.age.sex (merged Step 2 simset)



Nigeria: hiv.mortality.by.age (merged Step 2 simset)



Nigeria: hiv.mortality.by.age.sex (merged Step 2 simset)

